Q1: The Probability of Shooting at a Target

Problem Statement

A point is chosen at random from a disk of radius 10. Let A be the event that the point lies within 1 unit of the boundary.

1. Model the experiment assuming the point is uniformly distributed with respect to area, and compute $P(A)$.
2. Model the experiment assuming the distance to the point from the center is chosen uniformly from $[0, 10]$, and the direction is chosen independently and uniformly from $[0, 2\pi)$ and compute $P(A)$.
3. Explain why the two answers differ.

Solution

Part 1: Uniform distribution with respect to area

The disk has radius $R = 10$.

Total area:

$$A_{\text{total}} = \pi R^2 = \pi \cdot 10^2 = 100\pi.$$

Event A: point lies within 1 unit of the boundary. This means the point is outside a central disk of radius 9.

Area within 1 unit of the boundary:

$$A_A = \pi R^2 - \pi \cdot 9^2 = \pi(100 - 81) = 19\pi.$$

Therefore:

$$P(A) = \frac{19\pi}{100\pi} = \frac{19}{100} = 0.19.$$

Part 2: Uniform distance from center, uniform direction

Now the radius R (distance from center) is chosen uniformly in $[0, 10]$, so the distribution of R is uniform over length, not area. Direction Θ is uniform in $[0, 2\pi)$ and independent of R.

Event A occurs when the point is within 1 unit of the boundary, i.e., when $R > 9$.

Since $R \sim \text{Uniform}[0, 10]$:

$$P(A) = \frac{10 - 9}{10} = \frac{1}{10} = 0.1.$$

Part 3: Why the two answers differ

The two methods give different probabilities because they assume different underlying distributions for the random point.

- **Method 1 (uniform by area):** The probability density in polar coordinates is $f(r, \theta) = \frac{r}{50\pi}$ (since area element $dA = r\,dr\,d\theta$). This gives more weight to larger radii because area increases with r.
- **Method 2 (uniform radius, uniform direction):** The probability density is $f(r, \theta) = \frac{1}{20\pi}$ (constant in r), which means each radial distance is equally likely regardless of how much area it represents.

Mathematically:

$$P(\text{Boundary region}) \text{ (area method)} = \frac{\int_9^{10} r\,dr}{\int_0^{10} r\,dr} = \frac{(10^2 - 9^2)/2}{(10^2)/2} = \frac{100-81}{100} = 0.19.$$

$$P(\text{Boundary region}) \text{ (radius uniform)} = \frac{\int_9^{10} 1\,dr}{\int_0^{10} 1\,dr} = \frac{1}{10} = 0.1.$$

The second method **underestimates** the probability of being near the boundary because it fails to account for the larger area available at larger radii.

Final Answers

$$\boxed{0.19} \quad \text{(uniform by area)}$$

$$\boxed{0.1} \quad \text{(uniform radius, uniform direction)}$$